

Practical 6 - Analysis strategies for observational studies

Statistical Models for Discrete Outcomes

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Introduction

This practical is in two parts. In Part A you will use DAGs to inform selection of variables to controlled for in two example causal investigations. In Part B you will implement four variable selection methods used in investigations of causal effects, and compare the results: the full model approach, stepwise selection, change-in-estimates and double-selection.

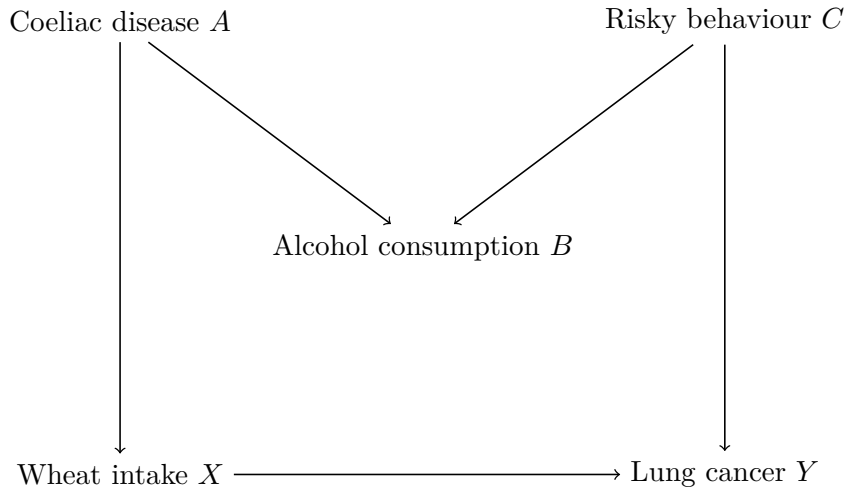
Part A: DAGs

Question 1

Consider the smoking and birth weight example from the lecture in Section 6.3.4, and the DAG in Figure 6.2. Suppose you wanted to estimate the causal effect of birth weight on survival. Which, if any, of the other variables in the DAG would you need to adjust for to do this?

Question 2

In another study it is of interest to estimate the causal effect of wheat intake on risk of lung cancer. The DAG is shown below, and all variables depicted in the DAG are measured in the observed data. Would you need to adjust for alcohol consumption to estimate the causal effect of wheat intake on lung cancer?



Part B: Variable selection strategies

In this part we will use a data set containing information on the birth weight of 4642 babies, alongside information about the baby's mother and her maternal history, and some information about the baby's father. The aim is to estimate the causal effect of maternal smoking (yes or no) on the risk of low birth weight, defined as birth weight $\leq 2500\text{g}$. The data were described in a paper by Cattaneo 2010 (Journal of Econometrics 155: 138-154), and are available directly from Stata using `use https://www.stata-press.com/data/r16/cattaneo2`, `clear`. Load the data into R using

```
library(haven)
cattaneo2 <- read_dta("cattaneo2.dta")
cattaneo2 <- zap_labels(cattaneo2)
cattaneo2$prevnewborndeadth <- cattaneo2$deadkids
```

Question 1

Explore the dataset.

- Identify which variables correspond to the outcome Y , exposure X and set of potential confounders Z . Are there are variables you think should not be included in Z ?
- One issue to look for is where there are very few observations in a particular level of a categorical variable. If there are, we can merge categories to avoid running into problems during regression analysis. Are there any variables you would consider re-categorising in this data?

Question 2

Summarise the exposure and outcome. Investigate the association between the mother's smoking status and risk of low birth weight, without adjustment for any other variables.

Question 3

Fit an appropriate regression model for the outcome Y given X and the full set of potential confounders Z , and interpret the coefficient for X . What is the estimated marginal mean difference $E(Y|do(X = 1)) - E(Y|do(X = 0))$, and what assumptions do you make to obtain this? Interpret your results.

Question 4

For a model with p parameters, the Akaike Information Criterion (AIC) is $-2 \log(L) + 2p$. The AIC compares improvement in the log-likelihood when adding new variable(s) relative to the increase in the number of parameters, with models with smaller AIC preferred.

Use the following code to apply forwards selection by AIC to select confounders to adjust for. Then, using the set of variables identified, again use the standardisation method to obtain an estimate of the marginal mean difference, $E(Y|do(X = 1)) - E(Y|do(X = 0))$, and corresponding confidence interval.

```
#model with just mbsmoke
mbsmokeMod <- glm(lbweight~mbsmoke,
                  data=cattaneo2,
                  family="binomial")
step(mbsmokeMod,
     direction = "forward",
     scope = list(lower = ~mbsmoke,
                   upper = ~mbsmoke + mmarried + mhispc + fhispc +
                             foreign + alcohol + prevnewbornddeath +
                             mage + medu + fage + fedu +
                             monthslb + order + mrace + frace + birthmonth))
```

Question 5

Install the `chest` package and use the `chest_glm` function to perform the change-in-estimates method using the forwards selection approach, stopping when the change in the OR estimate is less than 10%.

What set of adjustment variables do you end up with? Using this set of variables what is the estimated marginal mean difference, $E(Y|do(X = 1)) - E(Y|do(X = 0))$, and corresponding confidence interval.

Question 6

Perform the double-selection approach, using the `step` function for selection. What set of adjustment variables do you end up with? Using this set of variables what is the estimated marginal mean difference, $E(Y|do(X = 1)) - E(Y|do(X = 0))$, and corresponding confidence interval. When estimating the latter, use a robust sandwich variance by specifying the `vcov` argument of `avg_comparisons` as `HC1`.

Question 7

If you have time, you can try the double-selection method based on lasso. Install the `hdm` package from CRAN, then use the `rlassoEffect` function to estimate the effect of `mbsmoke` on `lbweight`. Note that in the documentation for this function, ‘controls’ means confounders. The terminology difference is due to the method coming from the econometrics literature. To see which confounders were selected, examine the `selection.index` component of the object returned by `rlassoEffect`.

Question 8

Make a table presenting your results from the different methods and write an accompanying paragraph summarising the results. Include some explanation as to why you see different results from the different methods (if this is the case).